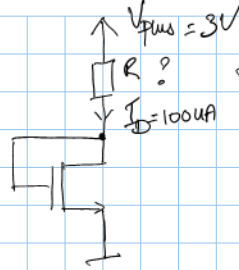
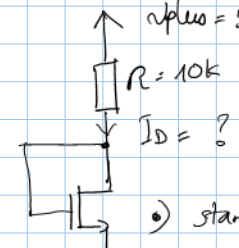


1)  $V_{plus} = 3V$
 $I_D = 100 \mu A$
 $I_D = 100 \mu A; V_{th} = 0,7V; k_n = 3,2 \text{ mA/V}^2$
 Da $V_{GS} = V_{DS} \Rightarrow$ Transistor ist im Abschnürrbereich
 $I_D = \frac{1}{2} k_n (V_{GS} - V_{th})^2$
 $V_{GS} = \pm \sqrt{\frac{2I_D}{k_n}} + V_{th} = \pm \sqrt{\frac{2 \cdot 100 \mu A}{3,2 \text{ mA/V}^2}} + 0,7V = \pm 0,25V + 0,7V$
 $V_{GS} = -0,25V + 0,7V = 0,45V$
 oder $V_{GS} = +0,25 + 0,7V = 0,95V$

Da $V_{GS} > V_{th} \Rightarrow V_{GS} = 0,95V$
 Maschengleichung: $V_{plus} = R \cdot I_D + V_{GS}$
 $R = \frac{V_{plus} - V_{GS}}{I_D} = 20,5 \text{ k}\Omega$

2)  $V_{plus} = 5V$
 $R = 10 \text{ k}\Omega$
 $V_{th} = 0,8V; k_n = 4 \text{ mA/V}^2$
 Iterationsverfahren
 $I_D(n) = \frac{V_{plus} - V_{GS}(n)}{R}$
 $V_{GS}(n+1) = \sqrt{\frac{2 \cdot I_D(n)}{k_n}} + V_{th}$
 • Startwert $V_{GS}(0) = 0V$; $I_D(0) = \frac{V_{plus} - V_{GS}(0)}{R} = \frac{5V - 0V}{10 \text{ k}\Omega} = 500 \mu A$

• $V_{GS}(1) = \sqrt{\frac{2 \cdot I_D(0)}{k_n}} + V_{th} = \sqrt{\frac{2 \cdot 500 \mu A}{4 \text{ mA/V}^2}} + 0,8V = 1,3V$; $I_D(1) = \frac{V_{plus} - V_{GS}(1)}{R} = \frac{5V - 1,3V}{10 \text{ k}\Omega} = 370 \mu A$

Abbruchkriterium: $|V_{GS}(1) - V_{GS}(0)| = |1,3V - 0V| = 1,3V > \epsilon_V$

$|I_D(1) - I_D(0)| = |370 \mu A - 500 \mu A| = 130 \mu A > \epsilon_I$

• $V_{GS}(2) = \sqrt{\frac{2 \cdot 370 \mu A}{4 \text{ mA/V}^2}} + 0,8V = 1,23012V$; $I_D(2) = \frac{5V - 1,23012V}{10 \text{ k}\Omega} = 376,988 \mu A$

Abbruchkriterium: $|1,23012V - 1,3V| = 69,88 \text{ mV} > \epsilon_V$

$|376,988 \mu A - 370 \mu A| = 6,988 \mu A > \epsilon_I$

• $V_{GS}(3) = \sqrt{\frac{2 \cdot 376,988 \mu A}{4 \text{ mA/V}^2}} + 0,8V = 1,23416V$; $I_D(3) = \frac{5V - 1,23416V}{10 \text{ k}\Omega} = 376,584 \mu A$

Abbruchkriterium: $|1,23416V - 1,23012V| = 4,04 \text{ mV} > \epsilon_V$

$|376,584 \mu A - 376,988 \mu A| = 404,298 \text{ nA} > \epsilon_I$

• $V_{GS}(4) = \sqrt{\frac{2 \cdot 376,584 \mu A}{4 \text{ mA/V}^2}} + 0,8V = 1,23393V$; $I_D(4) = \frac{5V - 1,23393V}{10 \text{ k}\Omega} = 376,607 \mu A$

Abbruchkriterium: $|1,23393V - 1,23416V| = 232,86 \text{ nV} > \epsilon_V$

$|376,607 \mu A - 376,584 \mu A| = 23,286 \text{ nA} < \epsilon_I$ ✓

• $V_{GS}(5) = \sqrt{\frac{2 \cdot 376,607 \mu A}{4 \text{ mA/V}^2}} + 0,8V = 1,23394V$; $I_D(5) = \frac{5V - 1,23394V}{10 \text{ k}\Omega} = 376,606 \mu A$

Abbruchkriterium: $|1,23394V - 1,23393V| = 13,4 \text{ nV} < \epsilon_V$ ✓

$|376,606 \mu A - 376,607 \mu A| = 1,34 \text{ nA} < \epsilon_I$ ✓

$\Rightarrow$ Antwort $I_D = I_{IS} = 376,606 \mu A$

Direkte Lösung der Quadratischen Gleichung

$I_D = \frac{1}{2} k_n (V_{GS} - V_{th})^2 = \frac{V_{plus} - V_{GS}}{R}$

$\frac{1}{2} k_n \cdot V_{GS}^2 + (-k_n \cdot V_{th} + \frac{1}{R}) \cdot V_{GS} + \frac{1}{2} k_n V_{th}^2 - \frac{V_{plus}}{R} = 0$

$\frac{1}{2} \cdot 4 \text{ mA} \cdot V_{GS}^2 + (-4 \text{ mA} \cdot 0,8 + \frac{1}{10 \text{ k}\Omega}) V_{GS} + \frac{1}{2} \cdot 4 \text{ mA} \cdot 0,8^2 - \frac{5}{10 \text{ k}\Omega} = 0$

$2 \text{ mA} \cdot V_{GS}^2 - 3,1 \text{ mA} \cdot V_{GS} + 0,78 \text{ mA} = 0$

$V_{GS1,2} = \frac{3,1 \pm \sqrt{3,1^2 - 4 \cdot 2 \cdot 0,78}}{2 \cdot 2}$

$V_{GS} = 1,2339V$ ✓ (weil $> V_{th}$) oder $V_{GS} = 0,71606V$ ✗ (weil $< V_{th}$)

$I_D = \frac{V_{plus} - V_{GS}}{R} = \frac{5V - 1,2339V}{10 \text{ k}\Omega} = 376,61 \mu A$

3)

$V_{th} = 0,7V$; $\mu_n C_{ox} = 200 \mu A/V^2$; $V_t = 1,5V$
 Da $I_g = 0 \Rightarrow V_{GS} = \frac{R_{G2}}{R_{G1} + R_{G2}} V_{pluo} = 1,5V$
 Da $V_{DS} = 1,5V > V_{GS} - V_{th} = 1,5 - 0,7V = 0,8V \Rightarrow$ Transistor ist im Abschnürbereich
 $I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 = \frac{V_{pluo} - V_D}{R_D}$
 $\frac{W}{L} = \frac{2 \cdot (V_{pluo} - V_D)}{R_D \cdot \mu_n C_{ox} \cdot (V_{GS} - V_{th})^2} = \underline{\underline{23,44}}$

Wenn R_D von $1k\Omega$ auf $2k\Omega$ geändert wird:

Annahme: Transistor ist im Abschnürbereich $\Rightarrow I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 = 1,5mA$
 $\Rightarrow V_{DS} = V_{pluo} - R_D \cdot I_D = 0V < V_{GS} - V_{th}$ $\checkmark$
 Annahme ist falsch $\Rightarrow$ Transistor ist im Triodenbereich $\checkmark$

$I_D = \frac{V_{pluo} - V_{DS}}{R_D} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} [2 \cdot (V_{GS} - V_{th}) V_{DS} - V_{DS}^2]$
 $\Leftrightarrow \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{DS}^2 - \left(\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th}) + \frac{1}{R_D} \right) V_{DS} + \frac{V_{pluo}}{R_D} = 0$
 $\frac{1}{2} \cdot 200 \mu \cdot 23,44 \cdot V_{DS}^2 - \left(200 \mu \cdot 23,44 (1,5 - 0,7) + \frac{1}{2k} \right) V_{DS} + \frac{3}{2k} = 0$
 $2,344 mA \cdot V_{DS}^2 - 4,2504 mA \cdot V_{DS} + 1,5 mA = 0$
 $V_{DS1,2} = \frac{4,2504 \pm \sqrt{4,2504^2 - 4 \cdot 2,344 \cdot 1,5}}{2 \cdot 2,344}$
 $V_{DS1,2} = 1,3334V \quad \text{oder} \quad 0,47993V$
 $\quad \quad \quad \hookrightarrow \text{Abschnürbereich} \quad \quad \quad \hookrightarrow \text{Triodenbereich}$

4)

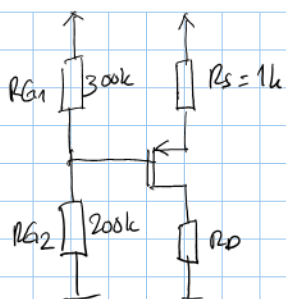
$V_{th} = 0,8V$; $\mu_n C_{ox} = 100 \mu A/V^2$; $W/L = 32$
 $I_D = 1mA$
 $V_D = V^+ - R_D \cdot I_D$
 $V_S = V^- + R_S \cdot I_D$
 $V_{GS} = V_G - V_S = -V^- - R_S \cdot I_D$

Annahme: Transistor ist im Abschnürbereich

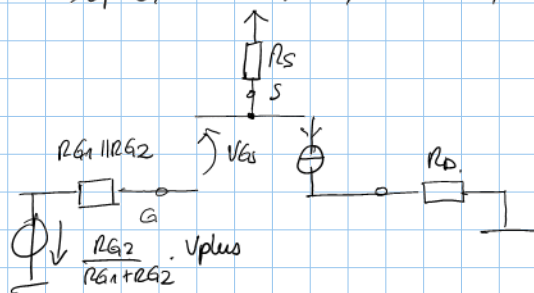
$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$
 $I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (-V^- - R_S \cdot I_D - V_{th})^2 = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V^- + R_S \cdot I_D + V_{th})^2$
 $I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} ((V^- + V_{th})^2 + 2(V^- + V_{th}) \cdot R_S \cdot I_D + R_S^2 I_D^2)$
 $0 = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \cdot R_S^2 \cdot I_D^2 + \left(\mu_n C_{ox} \frac{W}{L} \cdot (V^- + V_{th}) \cdot R_S - 1 \right) I_D + \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \cdot (V^- + V_{th})^2$
 $\Leftrightarrow \frac{1}{2} \cdot 100 \mu \cdot 32 \cdot 1k^2 I_D^2 + (100 \mu \cdot 32 \cdot (-5 + 0,8) \cdot 1k - 1) I_D + \frac{1}{2} \cdot 100 \mu \cdot 32 \cdot (-5 + 0,8)^2 = 0$
 $1600 I_D^2 - 14,4 I_D + 0,028224 = 0$
 $I_D = \frac{14,4 \pm \sqrt{14,4^2 - 4 \cdot 1600 \cdot 0,028224}}{2 \cdot 1600}$

1) $I_D = 6,11 mA \Rightarrow V_{DS} = V^+ - 1k \cdot 6,11 mA - V^- - 1k \cdot 6,11 mA = -2,23V$ $\times$
 2) $I_D = 2,8845 mA \Rightarrow V_{DS} = 4,23V$ $\checkmark \Rightarrow$ Transistor ist im Abschnürbereich

5)



$$\mu_p C_{ox} = 100 \mu A/V^2; W = 40 \mu; L = 1 \mu \quad V_{th} = -0,8V$$



$$V_{GS} = V_G - V_S = \frac{200k}{200k+300k} \cdot 5V - (V_{plus} - R_S \cdot I_D)$$

$$V_{GS} = 2V - 5V + R_S \cdot I_D = -3V + R_S \cdot I_D$$

Annahme: Transistor im Abschnürbereich:

$$I_D = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (-3V + R_S \cdot I_D - (-0,8))^2$$

$$\Leftrightarrow \frac{1}{2} \mu_p C_{ox} \frac{W}{L} \cdot R_S^2 \cdot I_D^2 - (2,2 \mu_p C_{ox} \frac{W}{L} \cdot R_S + 1) I_D + \frac{2,2^2}{2} \mu_p C_{ox} \frac{W}{L} = 0$$

$$\frac{1}{2} \cdot 100 \mu \cdot 40 \cdot 1k^2 \cdot I_D^2 - (2,2 \cdot 100 \mu \cdot 40 \cdot 1k + 1) I_D + \frac{2,2^2}{2} \cdot 100 \mu \cdot 40 = 0$$

$$2000 I_D^2 - 9,8 I_D + 0,00968 = 0$$

$$I_{D1,2} = \frac{+ 13 \pm \sqrt{13^2 - 4 \cdot 2000 \cdot 0,018}}{2 \cdot 2000}$$

$$I_D = 3,5282 \text{ mA} \Rightarrow V_{GS} = -3V + 1k \cdot 3,5282 \text{ mA} = 0,528 \text{ V} > V_{th} \quad \text{X}$$

$$I_D = 1,3718 \text{ mA} \Rightarrow V_{GS} = -3V + 1k \cdot 1,3718 \text{ mA} = -1,6282 \text{ V} < V_{th} \quad \checkmark$$

$$V_{DS} = V_D - V_S = R_D \cdot I_D - V_{plus} + R_S \cdot I_D = (R_D + R_S) I_D - V_{plus}$$

$$V_{DS} = (1k + 2,5k) \cdot 1,3718 \text{ mA} - 5V = -0,19870V$$

$$V_{GS} - V_{th} = -1,6282V - (-0,8V) = -0,8282V$$

 $\Rightarrow V_{DS} > V_{GS} - V_{th} \Rightarrow \text{Transistor ist eher im Triodenbereich} \Rightarrow \text{Annahme ist falsch.}$

Transistor im Triodenbereich

$$I_D = \mu_p C_{ox} \frac{W}{L} ((V_{GS} - V_{th}) V_{DS} - \frac{1}{2} V_{DS}^2)$$

$$I_D = \mu_p C_{ox} \frac{W}{L} ((-3V + R_S \cdot I_D - (-0,8)) \cdot ((R_D + R_S) \cdot I_D - V_{plus}) - \frac{1}{2} ((R_D + R_S) \cdot I_D - V_{plus})^2)$$

$$I_D = 100 \mu \cdot 40 \cdot ((-2,2 + 1k \cdot I_D) (3,5k \cdot I_D - 5) - \frac{1}{2} (3,5k \cdot I_D - 5)^2)$$

$$I_D = 4m (-7700 I_D + 11 + 3,5 \cdot 10^6 I_D^2 - 5000 I_D) - \frac{1}{2} \cdot 4m (12,25 \cdot 10^6 I_D^2 - 35k \cdot I_D + 25)$$

$$\Leftrightarrow 14000 I_D^2 - 50,8 I_D + 0,044 - 24500 I_D^2 + 70 I_D - 0,05 - I_D = 0$$

$$\Leftrightarrow 10500 I_D^2 - 18,2 I_D + 0,006 = 0$$

$$I_D = \frac{18,2 \pm \sqrt{18,2^2 - 4 \cdot 10500 \cdot 0,006}}{2 \cdot 10500}$$

$$1) I_D = 1,2906 \text{ mA} \Rightarrow V_{GS} = -3V + 1k \cdot 1,2906 \text{ mA} = -1,7094 < V_{th} = -0,8V$$

$$V_{DS} = (1k + 2,5k) \cdot 1,2906 \text{ mA} - 5V = -0,4829V$$

$$V_{GS} - V_{th} = -1,7094 - (-0,8V) = -0,9094V$$

 $\Rightarrow V_{DS} > V_{GS} - V_{th} \Rightarrow \text{Transistor ist im Triodenbereich} \quad \checkmark$

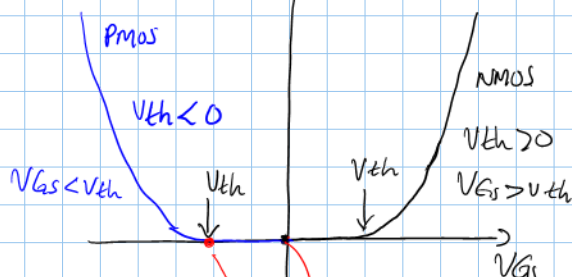
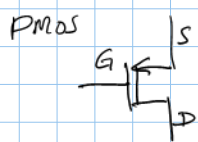
$$2) I_D = 0,443 \text{ mA} \Rightarrow V_{GS} = -3V + 1k \cdot 0,443 \text{ mA} = -2,5572V < V_{th}$$

$$V_{GS} - V_{th} = -2,5572V - (-0,8V) = -1,7572V$$

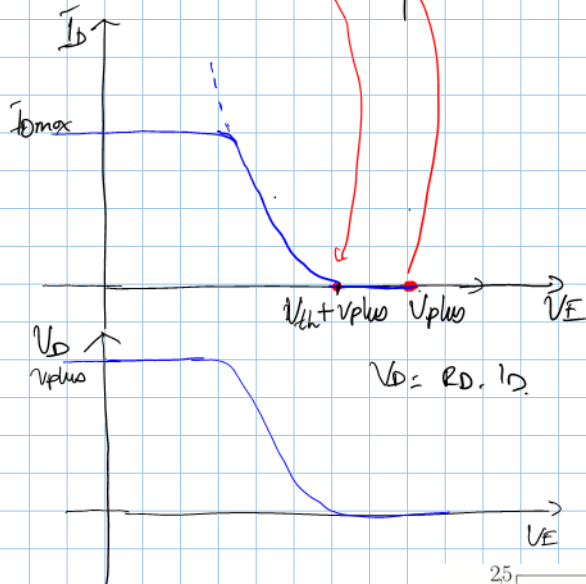
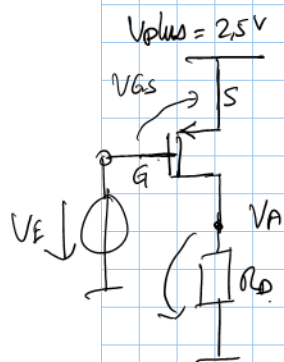
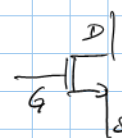
$$V_{DS} = (1k + 2,5k) \cdot 0,443 \text{ mA} - 5V = -3,4503V$$

 $\text{Da } V_{DS} < V_{GS} - V_{th} \Rightarrow \text{Transistor ist im Abschnürbereich} \quad \text{X}$
Die richtige Lösung ist $I_D = 1,2906 \text{ mA}$ und der Transistor ist im Triodenbereich betrieben.

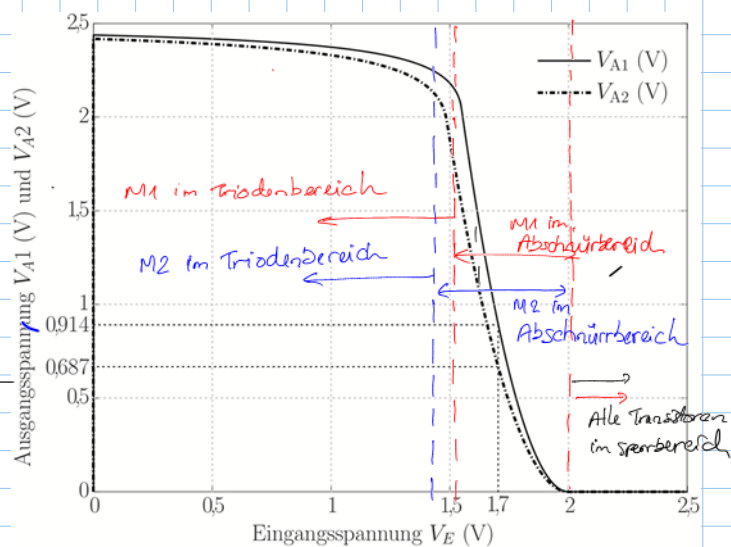
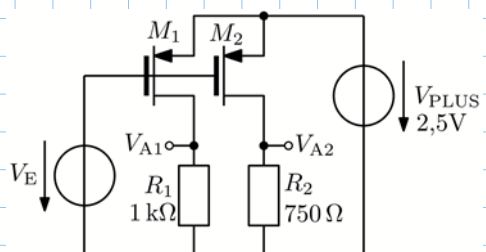
6) Grundlage - Eingangskennlinien $I_D - V_{GS}$ von PMOS & NMOS zum Vergleich



NMOS.



$$\begin{aligned} V_{GS} &= V_G - V_S = V_E - V_{plus} \\ V_E &= V_{GS} + V_{plus} \\ V_{GS} &= 0 \Rightarrow V_E = V_{plus} = 2,5 \\ V_{GS} &= V_{th} \Rightarrow V_E = V_p + V_{plus} \\ I_{Dmax} &= \frac{V_{plus}}{R_D} \end{aligned}$$



a) Die Kennlinien zeigen ein invertierendes Verhalten, deshalb kann es sich nur um PMOS-Transistoren in Source-Schaltung handeln.

b) Siehe Diagramm.

c) Aus dem Diagramm: $V_E = V_G > 2V \Rightarrow$ Transistoren sperren.
 $V_{GS} + V_S > 2V$

$$V_{GS} > 2V - 2,5 = -0,5V \Rightarrow V_{GS} < -0,5V \Rightarrow \text{Transistoren leiten.}$$

$\Rightarrow$ Schwellenspannung $V_{th} = -0,5V$

d) $V_E = V_G = V_{GS} + V_S = 1,7V \Rightarrow V_{GS} = 1,7V - 2,5V = -0,8V$

$$V_{D1} = 914mV \Rightarrow I_{D1} = V_{D1}/R_1 = 914mV/1k\Omega = 914\mu A$$

$$I_{D1} = \frac{1}{2} k_p (V_{GS} - V_{th})^2 (1 + \lambda V_{DS1}) \approx 914\mu A = \frac{1}{2} k_p (-0,8 - (-0,5))^2 (1 + \lambda (914mV - 2,5V))$$

$$914\mu A = 0,045k_p (1 - 1,586\lambda) \quad \dots \text{G.1}$$

$$V_{D2} = 687mV \Rightarrow I_{D2} = V_{D2}/R_2 = 687mV/750\Omega = 916\mu A$$

$$I_{D2} = \frac{1}{2} k_p (V_{GS} - V_{th})^2 (1 + \lambda V_{DS2}) = 916\mu A = \frac{1}{2} k_p (-0,8 - (-0,5))^2 (1 + \lambda (687mV - 2,5V))$$

$$916\mu A = 0,045k_p (1 - 1,813\lambda) \quad \dots \text{G.2}$$

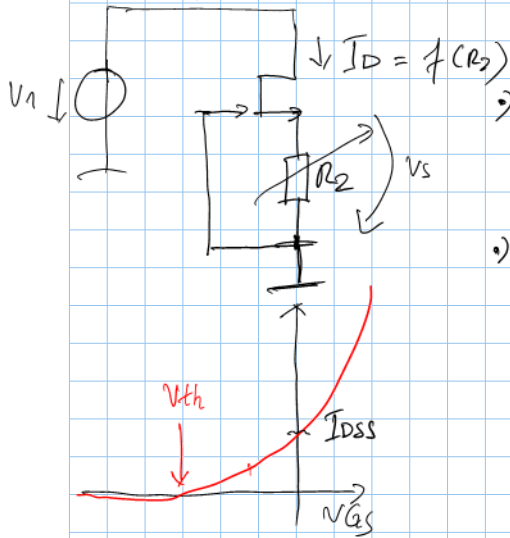
$$G_2/G_1 \Rightarrow 1,0022 = \frac{(1 - 1,813\lambda)}{(1 - 1,586\lambda)} \Leftrightarrow 1,0022 - 1,5895\lambda = 1 - 1,813\lambda$$

$$0,0022 = -0,2235\lambda$$

$$\lambda = -0,01 \quad 1/V$$

$$914\mu A = 0,045k_p (1 - 1,586 \cdot (-0,01)) \Rightarrow k_p = 20 \text{ mA/V}^2$$

JFET-Transistor als Stromquelle.



$$I_D = \frac{1}{2} k_n (V_{GS} - V_{th})^2 \quad \text{im Abschnürbereich.}$$

$$R_2 = 500 \Omega \Rightarrow I_D = 2 \text{ mA} =$$

$$V_S = R_2 \cdot I_D = 500 \Omega \cdot 2 \text{ mA} = 1 \text{ V}$$

$$V_{GS} = V_G - V_S = 0 - 1 \text{ V} = -1 \text{ V}$$

$$\Rightarrow 2 \text{ m} = \frac{1}{2} k_n (-1 \text{ V} - V_{th})^2 \quad \dots \quad 1)$$

$$R_2 = 4 \text{ k} \Omega \Rightarrow I_D = 0,5 \text{ mA}$$

$$V_S = R_2 \cdot I_D = 4 \text{ k} \cdot 0,5 \text{ m} = 2 \text{ V} \Rightarrow V_{GS} = -2 \text{ V}$$

$$\Rightarrow 0,5 \text{ m} = \frac{1}{2} k_n (-2 \text{ V} - V_{th})^2 \quad \dots \quad 2)$$

$$\frac{(1)}{(2)} = \pm \sqrt{4} = \frac{(-1 \text{ V} - V_{th})}{(-2 \text{ V} - V_{th})}$$

$$\pm 2 = \frac{-1 \text{ V} - V_{th}}{-2 \text{ V} - V_{th}}$$

$$(+)\ 2(-2 \text{ V} - V_{th}) = -1 \text{ V} - V_{th}$$

$$\boxed{-3 \text{ V} = V_{th}}$$

$$\left. \begin{aligned} -2(-2 \text{ V} - V_{th}) &= -1 \text{ V} - V_{th} \\ 5 V_{th} &= -5 \text{ V} \\ V_{th} &= -1,667 \text{ V} \end{aligned} \right\}$$

$$5 V_{th} = -5 \text{ V}$$

$$V_{th} = -1,667 \text{ V}$$

$$(1) \quad 2 \text{ m} = \frac{1}{2} k_n (-1 \text{ V} - (-3 \text{ V}))^2$$

$$k_n = \frac{2 \cdot 2 \text{ mA}}{(2 \text{ V})^2} = 1 \text{ mA/V}^2$$

$$I_{DSS} = \frac{1}{2} k_n (0 - V_{th})^2 = \frac{1}{2} \cdot 1 \text{ mA/V}^2 \cdot (-(-3 \text{ V}))^2 = 4,5 \text{ mA}$$

b) Abschnürbereich $V_{DS} \geq V_{GS} - V_{th}$ $V_{GS} = V_G - V_S = -V_S$

$$V_1 = V_{DS} + V_S = V_{DS} - V_{GS} \Rightarrow V_{DS} = V_1 + V_{GS}$$

$$V_1 + V_{GS} \geq V_{GS} - V_{th}$$

$$V_1 \geq -V_{th} \Rightarrow \underline{V_1 \geq 3 \text{ V}}$$

c) $R_2 = 1 \text{ k} \Omega \Rightarrow I_D = ?$ $V_{GS} = 0 - R_2 \cdot I_D$

$$I_D = \frac{1}{2} k_n (V_{GS} - V_{th})^2 = \frac{1}{2} k_n (0 - R_2 \cdot I_D - V_{th})^2$$

$$I_D = \frac{1}{2} k_n R_2^2 \cdot I_D^2 + k_n R_2 \cdot V_{th} \cdot I_D + \frac{1}{2} k_n V_{th}^2$$

$$0 = \frac{1}{2} k_n R_2^2 \cdot I_D^2 + (k_n R_2 \cdot V_{th} - 1) I_D + \frac{1}{2} k_n V_{th}^2$$

$$0 = a \cdot I_D^2 + b \cdot I_D + c \quad \leftarrow$$

$$I_{D1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = \frac{1}{2} k_n R_2^2 = \frac{1}{2} 1 \text{ m} \cdot 1 \text{ M} = 500$$

$$b = k_n R_2 \cdot V_{th} - 1 = 1 \text{ m} \cdot 1 \text{ k} \cdot (-3) - 1 = -4$$

$$c = \frac{1}{2} k_n V_{th}^2 = \frac{1}{2} 1 \text{ m} \cdot (-3)^2 = 4,5 \text{ m}$$

$$I_{D1,2} = \frac{-(-4) \pm \sqrt{16 - 4 \cdot 500 \cdot 4,5 \text{ m}}}{2 \cdot 500} = \frac{4 \pm 2,65}{1000}$$

$$I_{D1} = 6,65 \text{ mA}$$

oder

$$I_{D1} = 1,35 \text{ mA}$$

$$\downarrow$$

$$V_{GS} = 0 - 6,65 \text{ m} \cdot 1 \text{ k} = -6,65 \text{ V}$$

$$\downarrow$$

$$V_{GS} = 0 - 1,35 \text{ m} \cdot 1 \text{ k} = -1,35 \text{ V}$$

$V_{GS} < V_{th}$ nicht Abschnürbereich

$V_{GS} > V_{th} \Rightarrow$ Abschnürbereich

d) $R_1 = 1k\Omega$ $V_1 = 2V$ $\Rightarrow$ da $V_1 < 3V \Rightarrow$ der Transistor im Triodenbereich.

$$I_D = k_n \left((V_{GS} - V_{th}) V_{DS} - \frac{1}{2} V_{DS}^2 \right)$$

$$\frac{V_1 - V_{DS}}{R_2} = k_n \left((V_{DS} - V_1 - V_{th}) V_{DS} - \frac{1}{2} V_{DS}^2 \right)$$

$$V_1 = V_{DS} + R_2 \cdot I_D$$

$$I_D = \frac{V_1 - V_{DS}}{R_2}$$

$$V_1 - V_{DS} = k_n R_2 V_{DS}^2 - k_n R_2 (V_1 + V_{th}) V_{DS} - \frac{1}{2} k_n R_1 V_{DS}^2$$

$$V_{DS} = V_1 + V_{GS}$$

$$0 = \frac{1}{2} k_n R_2 V_{DS}^2 + (1 - k_n R_2 (V_1 + V_{th})) V_{DS} - V_1$$

$$0 = a V_{DS}^2 + b V_{DS} + c$$

$$a = \frac{1}{2} k_n R_2 = \frac{1}{2} \cdot 1m \cdot 1k = 0,5$$

$$b = 1 - k_n R_2 (V_1 + V_{th}) = 1 - 1m \cdot 1k \cdot (2V - 3V) = 2$$

$$V_{DS1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$V_{DS1,2} = \frac{-2 \pm \sqrt{4 + 4}}{2 \cdot 0,5}$$

$$c = -V_1 = -2$$

$$V_{DS1} = 0,83V \checkmark$$

$\hookrightarrow V_{DS} > 0$

$$V_{DS2} = -4,83V \times$$